Healthcare Data Imputation

Abstract

Missing data is a persistent obstacle in healthcare machine learning, where clinical records are frequently incomplete due to irregular data collection, patient dropout, or measurement cost. This project investigates how missing data affects downstream prediction performance on a clinical dataset, compares four imputation methods (mean, median, k-nearest neighbours, and an iterative multivariate imputer), and evaluates whether Genetic Algorithm (GA)-based feature selection improves Random Forest prediction after imputation. Using the Heart Failure Clinical Records dataset (UCI, 299 patients, 12 features, target DEATH_EVENT), missingness was simulated under three mechanisms - Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR) - at 10%, 20%, and 30% rates, with ground truth preserved for every intentionally masked cell. Across a 180-run multi-seed matrix (3 mechanisms x 3 levels x 4 imputers x 5 seeds) and 27 paired statistical tests, missingness itself was found to significantly reduce prediction performance relative to complete data in 4 of 9 complete-vs-imputed comparisons ($p < 0.05$), while no imputation method or missingness mechanism was statistically distinguishable from any other at this sample size (0 of 18 comparisons significant). GA feature selection reliably and reproducibly selected a small, clinically plausible feature subset (time, ejection_fraction chosen in 5 of 5 independent runs) but did not reliably improve prediction accuracy once replicated across multiple seeds. These results are reported without adjustment to favour any predetermined conclusion: the project's central, statistically defensible finding is that the presence of missing data matters more than the specific mechanism or imputation method used to address it, and that GA's genuine contribution on this dataset is principled feature reduction rather than an accuracy gain.

1. Introduction

Electronic health records and clinical study datasets are rarely complete. Values go unrecorded because a test was not ordered, a patient missed a follow-up visit, or a field was left blank during data entry. How a predictive model handles this missingness - whether through discarding incomplete records, imputing plausible replacement values, or restricting itself to a robust subset of features - directly affects the reliability of any downstream clinical decision-support tool. This project studies that problem empirically on a single, well-characterised clinical dataset, using a controlled experimental design in which missingness is deliberately introduced so that the true values remain known and imputation error can be measured directly, rather than relying on a dataset's uncontrolled, unverifiable natural missingness.

2. Background

Three formal mechanisms describe how data can go missing (Rubin, 1976). Missing Completely at Random (MCAR) means the probability of a value being missing is unrelated to any observed or unobserved variable. Missing at Random (MAR) means the probability of missingness depends on other observed variables. Missing Not at Random (MNAR) means the probability of missingness depends on the (unobserved) value itself. These mechanisms have different practical implications: simple imputation methods that assume MCAR can introduce bias when the true mechanism is MAR or MNAR. Classical imputation approaches range from simple univariate statistics (mean/median substitution) to similarity-based methods (k-nearest neighbours) to fully multivariate, iterative approaches such as MICE (van Buuren & Groothuis-Oudshoorn, 2011) and MissForest (Stekhoven & Buhlmann, 2012). Feature selection via Genetic Algorithms has a long history in applied machine learning (Leardi, Boggia, & Terrile, 1992) as a population-based wrapper method that can search a combinatorial feature space more effectively than exhaustive or greedy approaches, at the cost of a stochastic, CV-fitness-driven search that can overfit small training sets.

3. Problem Statement

Two related problems are addressed. First, the effect of missing data on downstream prediction is often reported inconsistently across the literature because studies vary in dataset, missingness mechanism, missingness rate, and imputation method simultaneously, making it difficult to isolate which factor drives observed performance changes. Second, feature selection methods such as GA are frequently proposed as a way to both reduce dimensionality and improve accuracy, but claims of GA "improving" prediction are sometimes drawn from a single train/test split, which cannot distinguish a genuine effect from split-specific noise. This project addresses both problems using a single fixed dataset, a controlled missingness framework with known ground truth, and a repeated multi-seed design with paired statistical testing, so that any claimed effect - of missingness, of imputer choice, or of GA - can be checked for replication rather than accepted from a single run.

4. Objectives

1. Quantify how missingness (across mechanism and rate) affects Random Forest prediction performance relative to a complete-data baseline. 2. Compare four imputation methods (mean, median, KNN, IterativeImputer) on both reconstruction accuracy and downstream prediction performance. 3. Determine whether GA-based feature selection improves prediction performance, or preserves comparable performance with fewer features, both on complete data and after imputation. 4. Situate the project's own results against selected published work on imputation and GA-based feature selection in healthcare prediction, without presenting different-dataset results as direct head-to-head comparisons.

5. Research Questions

- RQ1. How does missing data affect healthcare prediction performance?
- RQ2. Which selected imputation method performs best for the chosen dataset?
- RQ3. Can GA-based feature selection improve prediction performance, or preserve comparable performance using fewer features?
- RQ4. How does Imputation + GA + Random Forest compare with selected baselines and existing research?

6. Literature Review

Fifteen references were identified and verified against primary or indexed bibliographic sources (results/literature_matrix.csv; full discussion in docs/literature_review.md). Rubin (1976) provides the foundational MCAR/MAR/MNAR taxonomy used throughout this project. Van Buuren & Groothuis-Oudshoorn (2011) and Stekhoven & Buhlmann (2012) describe the multivariate iterative imputation family that motivates this project's IterativeImputer method, while Troyanskaya et al. (2001) established KNN imputation in a biological data context. Breiman (2001) is the foundational Random Forest reference underlying the prediction model used throughout. Leardi, Boggia, & Terrile (1992) is a foundational reference for GA-based feature selection. Demsar (2006) informed this project's use of paired statistical tests rather than a single point-estimate comparison across methods.

Among recent clinical-imputation studies, Shadbahr et al. (2023) found, across several independent datasets, that the degree of missingness - not the specific mechanism or imputer - was the dominant driver of downstream classifier degradation; this project's own RQ1 result (Section 12) independently arrives at the same qualitative conclusion on a different dataset. Ren et al. (2024) systematically reviewed machine-learning-based imputation in electronic health records and reported no single imputer that consistently dominates across studies, consistent with this project's RQ2 finding. Aracri et al. (2025) compared missing-data imputation methods for a dementia classification task and likewise found no single method statistically dominant. Chicco & Jurman (2020), the origin study for this project's dataset, identified serum creatinine and ejection fraction as the two most predictive features for heart failure survival using an independent feature-importance analysis; this project's GA feature-selection result (Section 15) selects ejection_fraction in every run and serum_creatinine in 4 of 5 runs, an independent cross-check that agrees with the origin paper without having used its analysis. Kumar & Sahoo (2017) reported a favourable GA+Naive-Bayes result for a cardiovascular classification task; this project's own GA+Random-Forest result (Section 16) diverges from that finding, and this divergence is discussed directly rather than omitted (Section 20). Pudjihartono et al. (2022) reviewed feature-selection methods for disease-risk prediction and discusses hybrid filter-then-wrapper approaches as a way to reduce a wrapper method's overfitting risk on small training sets, which motivates this project's future-work discussion of GA overfitting (Section 21). Two entries - Kumar & Sahoo (2017) and Yaqoob et al. (2025) - carry a verification caveat: their full text could not be retrieved (HTTP 403 from the publisher), so bibliographic details and reported findings are drawn from indexed metadata only. Grzesiak et al. (2025) is cited as an unreviewed arXiv preprint and is labelled as such. Yaqoob et al. (2025) itself uses Seagull Optimization, not a genetic algorithm, and is cited only for general feature-selection context, not as a GA comparison point.

7. Dataset

The Heart Failure Clinical Records dataset (UCI Machine Learning Repository, ID 519, CC BY 4.0 license) was selected as the project's single primary dataset after comparison against four other candidates - Chronic Kidney Disease, Heart Disease (Cleveland), Indian Liver Patient Dataset, and Cervical Cancer Risk Factors - scored across twelve criteria including healthcare relevance, missing-data suitability, sample size, class balance, and licensing (results/dataset_candidate_comparison.csv, docs/dataset_selection.md). It contains 299 patient records, 12 features, and a binary target, DEATH_EVENT. The dataset is fully observed: zero missing values and zero duplicate rows were found on integrity checking, and its SHA-256 checksum and full provenance are recorded in data/raw/heart_failure_metadata.json. The class distribution is moderately imbalanced: 203 survivors (0) and 96 deaths (1), approximately 68%/32%.

This dataset was selected specifically because its complete, natural state allows the complete-data Random Forest baseline to use all 299 rows without any natural missing values confounding the analysis, and because every cell that is later intentionally masked for the controlled MCAR/MAR/MNAR experiments has a known, verifiable ground-truth value - the cleanest possible setup for measuring imputation error directly. Against this benefit, the dataset's twelve features are fewer than some rejected candidates (for example, Chronic Kidney Disease's twenty-four), which makes GA's absolute feature-count reduction look smaller in raw terms; this trade-off was judged acceptable given that a clean, unconfounded experimental design was prioritised over a larger but less controlled feature space.

One feature, time (days of follow-up before the recorded outcome), requires explicit methodological comment: in survival-style clinical data, a short follow-up interval correlates with the death event almost by construction, since patients who die are, by definition, not followed further. time was retained in the dataset, since removing it would itself be an undocumented methodological choice not called for by any experimental requirement, but its outsized influence on Random Forest performance is treated throughout this report as a modelling caveat rather than a genuine clinical discovery (see Section 20).

8. Preprocessing

Data loading, target detection, and configuration are centralised in src/data_loader.py and src/config.py. An 80/20 stratified train/test split is performed once per seed (train_test_split_stratified, src/preprocessing.py), using scikit-learn's stratified train_test_split with the seed passed directly as random_state; this means each of the project's five repeated seeds draws a genuinely fresh train/test partition, not merely a fresh source of model randomness. Stratified 5-fold cross-validation is used inside the training split wherever a fitness or tuning signal is required (GA fitness evaluation, in particular). The held-out test split at no point participates in imputer fitting, GA fitness evaluation, or any other model-selection decision, consistent with a strict train-then-transform discipline (Section 10).

9. Missing-Data Mechanisms

Three missingness mechanisms were implemented in src/missingness.py:

- MCAR: eligible cells are masked independently at random, with no dependence on any observed or unobserved variable.

- MAR: the probability of a cell being masked is conditioned on other observed variables, with the exact conditioning variables and direction documented per feature in docs/experiment_log.md.

- MNAR: the probability of a cell being masked is conditioned on the value being masked itself (or a defensible approximation), again with the exact assumption documented per feature.

All three mechanisms were applied at three target rates - 10%, 20%, and 30% - and scoped to the seven continuous features in the dataset; the five binary clinical flags were excluded from masking, since mean/median imputation is not a meaningful reconstruction target for a binary indicator. The target column and any identifier columns were never masked. For every intentionally masked cell, the true value was preserved before masking so that imputation error could later be measured only against genuinely hidden ground-truth values - no cell’s imputation error is ever computed against a naturally missing value, since this dataset has none. Realised missingness percentages closely matched the requested percentages throughout (results/tables/missingness_summary.csv); for example, the MCAR training-split realised rates were 10.04%, 20.08%, and 30.13% against requested 10/20/30%. A dedicated statistical sanity check confirmed that the MAR and MNAR masking probabilities actually correlated with their documented conditioning variables as designed (point-biserial correlation, $p < 0.05$).

10. Methodology

The experimental pipeline follows a fixed conceptual flow: raw data is loaded and validated, split into training and test partitions, missingness is introduced onto each partition independently, imputers are fit on the training partition only and then used to transform both partitions, GA feature selection (where used) operates exclusively on training-partition cross-validation, and the final Random Forest model is trained on the (possibly imputed, possibly feature-selected) training data and evaluated once on the untouched held-out test partition. This split-before-fit discipline was re-verified directly in the project’s final experiment audit by re-reading the relevant source code (src/preprocessing.py, src/genetic_selection.py) and confirming that no fitting function accepts or references test-split data; a dedicated automated leakage test additionally asserts that GA fitness evaluation only ever receives training-split row indices (tests/test_run_e6.py). No hyperparameter or method-selection decision was ever informed by the held-out test set.

11. Imputation Methods

Four imputation methods were implemented in src/imputation.py:

- Mean imputation: replaces a missing value with the training-split mean of its column.

- Median imputation: replaces a missing value with the training-split median of its column.

- K-nearest neighbours (KNN) imputation: imputes a missing value from the average of its k = 5 nearest neighbours (scikit-learn default), fit on the training split.

- IterativeImputer: scikit-learn's round-robin multivariate imputer, used here as a single-imputation, computationally lighter alternative to a full multiple-imputation procedure such as MICE or MissForest - a deliberate scope-control choice appropriate to this project's dataset size and single-semester timeframe (docs/literature_review.md).

An optional Autoencoder-based imputer (permitted by the project's execution plan as an optional method) was not implemented: at 12 features and 299 rows, an autoencoder is not scientifically justified relative to the classical and iterative methods already implemented, which are sufficient to answer RQ1 and RQ2 on this dataset's scale.

12. GA Feature Selection

Genetic Algorithm feature selection was implemented in src/genetic_selection.py. Each candidate feature subset is represented as a binary chromosome (1 = feature selected, 0 = feature removed), with the all-zero chromosome disallowed. The GA used tournament selection, single-point crossover, and elitism, with the following parameters, matching the project's specified defaults exactly: population size 30, 30 generations, crossover probability 0.8, tournament size 3, elite count 2, and mutation probability approximately 1/12 (the reciprocal of the twelve-feature chromosome length). Fitness was defined as the mean 5-fold stratified cross-validation F1 score, computed on the training split only, minus a small feature-count penalty of 0.02 multiplied by the selected-feature ratio, discouraging unnecessarily large subsets without dominating the accuracy term. Fitness evaluation used 50 trees per Random Forest fit rather than the final model's 200, a documented cost-reduction choice justified by the observation that CV-F1 ranking between candidate subsets is stable well before 200 trees, and because fitness is evaluated up to 900 times in a single GA run; the final all-features-versus-GA comparison uses the full 200-tree configuration on both sides, so this shortcut affects only the search process and not the reported comparison numbers. The nine-cell imputation-plus-GA sweep (E6) used a reduced population of 20 and 15 generations for compute-budget reasons; this reduction is documented in src/config.py and did not apply to the complete-data GA runs (E5).

13. Random Forest

Random Forest (scikit-learn) was used as the sole downstream prediction model throughout the project, per the project's scope-control decision to keep the study to one primary classifier. All final reported comparisons use 200 trees. Accuracy, Precision, Recall, F1-score, and ROC-AUC were recorded for every run; hyperparameters, seed, and runtime were logged for every result. No hyperparameter was ever tuned against the held-out test set.

14. Experimental Setup

The project's experiment groups (Section 28 of the governing execution plan) are summarised below, all implemented, executed, and validated:

Group	Description	Status
E0	Complete-data Random Forest baseline	Done single seed and 5-seed repeat
E1	Mean/median imputation	Done MCAR/MAR/MNAR x 10/20/30%, multi-seed
E2	KNN imputation	Done same coverage as E1
E3	IterativeImputer (ML-based) imputation	Done same coverage as E1
E4	Autoencoder (optional)	Not attempted not scientifically justified at this dataset's scale
E5	GA feature selection	Done 5 seeds on complete data
E6	Imputation + GA + Random Forest	Done full 9-cell single-seed sweep, plus a 3-cell/3-seed multi-seed replicat
E7	Literature comparison	Done contextual only, no cross-dataset head-to-head claims

The multi-seed missingness matrix covers 3 mechanisms (MCAR/MAR/MNAR) x 3 levels (10/20/30%) x 4 imputers (mean/median/KNN/iterative) x 5 seeds, for 180 total runs (results/metrics/e1_e2_e3_multiseed.csv, aggregated in e1_e2_e3_multiseed_summary.csv). The GA matrix consists of 5 independent complete-data runs (E5, one per seed) and a 9-cell single-seed sweep after imputation (E6), with the three most informative E6 cells (largest apparent GA gain, largest apparent GA loss, and a near-tie at the single seed) replicated across 3 seeds (results/metrics/e6_multiseed_subset.csv). Full 9-cell/5-seed E6 coverage was not pursued, for compute-budget reasons, since the 3-cell subset already demonstrates a consistent pattern (Section 16); this is documented as an accepted, non-blocking limitation rather than a silent omission. Five seeds - 42, 123, 2026, 7, and 99 - were used throughout, matching the project's specified defaults. Rather than a single master run_all.py script, the project uses one independently re-runnable script per experiment group under experiments/, a deliberate scope-control simplification for a single-dataset, workstation-scale project, documented rather than silently deviating from the suggested execution pattern.

15. Metrics

Imputation quality was measured on intentionally masked cells only, using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) per feature, mechanism, missingness level, and seed. Prediction quality was measured using Accuracy, Precision, Recall, F1-score, and ROC-AUC on the held-out test split. Given the wide difference in feature scales (for example, platelets is on the order of 10^4 while serum_creatinine is on the order of 1), per-feature MAE/RMSE values are reported individually rather than averaged across features, since a single cross-feature average would not be analytically meaningful.

16. Results

Complete-data baseline (E0). Across 5 seeds, the complete-data Random Forest baseline achieved Accuracy 0.863 +/- 0.040, Precision 0.802 +/- 0.079, Recall 0.768 +/- 0.156, F1 0.776 +/- 0.081, and ROC-AUC 0.918 +/- 0.031 (results/metrics/repeated_seeds_e0.csv). The single-seed (seed 42) held-out result was F1 0.667, ROC-AUC 0.883 - the gap between this single-seed number and the 5-seed mean illustrates directly why no single-seed result is treated as a final claim anywhere in this project.

Imputation reconstruction quality. Mean, median, and IterativeImputer performed similarly to one another at every missingness level (results/metrics/e1_mean_median_imputation.csv, e3_iterative_imputation.csv). KNN (k = 5, untuned) was consistently the worst reconstructor by MAE/RMSE across this dataset's feature scales (results/metrics/e2_knn_imputation.csv).

Prediction after imputation, multi-seed. Across the full 180-run multi-seed matrix (results/metrics/e1_e2_e3_multiseed_summary.csv), F1 ranged approximately from 0.615 to 0.742 across the nine mechanism-by-level cells, degrading as missingness rose from 10% to 30%, as expected. Reconstruction quality and downstream prediction quality did not track one another: KNN, the worst reconstructor by MAE/RMSE, was sometimes competitive with or better than the other imputers on downstream F1 - a disconnect discussed further in Section 20.

Statistical tests. Twenty-seven paired comparisons were run across the 5-seed matrix (paired t-test with a Wilcoxon signed-rank cross-check; results/tables/statistical_tests.csv, full methodology in docs/statistical_analysis.md): 9 imputer-versus-imputer comparisons, 9 mechanism-versus-mechanism comparisons, and 9 complete-data-versus-imputed-data comparisons. Of these, 4 of 9 complete-versus-imputed comparisons were significant at $p < 0.05$ (MCAR 10%, MCAR 20%, MNAR 20%, MNAR 30%), while 0 of 9 imputer-versus-imputer and 0 of 9 mechanism-versus-mechanism comparisons reached significance. No multiple-comparison correction was applied, a documented choice: at $n = 5$ seeds, any standard correction procedure would push the significance threshold low enough that almost no comparison could ever reach it, regardless of true effect size; individual p-values are read as exploratory rather than confirmatory.

GA feature selection (E5). Across 5 independent GA runs on complete data, each converging to a

4-to-7-feature subset out of 12, time and ejection_fraction were selected in every run (frequency 1.0), serum_creatinine in 4 of 5 runs (0.8), diabetes in 3 of 5 (0.6), and age in 0 of 5 runs - a genuine negative finding, not an omission (results/tables/ga_feature_frequency.csv, figures/ga_feature_frequency.png, figures/ga_convergence.png).

All-features versus GA-selected (complete data, seed 42). Using the majority-vote subset (features selected in at least 50% of runs: time, ejection_fraction, serum_creatinine, diabetes - a 4-of-12, 66.7% reduction), Accuracy improved from 0.817 to 0.833 and F1 from 0.667 to 0.706, while ROC-AUC fell from 0.883 to 0.815 (results/metrics/all_features_vs_ga_complete.csv) - a mixed single-seed result, not presented as an unqualified GA win.

Imputation + GA + Random Forest (E6). Across the nine mechanism-by-level cells at seed 42, GA improved F1 in 3 of 9 cells, matched it in 1 of 9, and reduced it in 5 of 9 (results/metrics/e6_imputed_ga_rf.csv, results/tables/four_way_comparison.csv, figures/four_way_comparison.png). The three most informative cells from this single-seed sweep - the largest apparent GA gain (MCAR 30%), the largest apparent GA loss (MNAR 30%), and a near-tie (MAR 20%) - were replicated across 3 seeds (results/metrics/e6_multiseed_subset.csv). Averaged over those 3 seeds, GA F1 was at or below all-features F1 in every one of the three cells (MCAR 30%: 0.632 versus 0.647; MAR 20%: 0.676 versus 0.724; MNAR 30%: 0.607 versus 0.634) - the single-seed "GA gain" at MCAR 30% did not replicate.

17. Statistical Analysis

The paired statistical testing strategy (Section 16; full detail in docs/statistical_analysis.md) used a paired t-test as the primary test, with a Wilcoxon signed-rank test run as a non-parametric cross-check for each of the 27 comparisons, since the same 5 seeds are paired across the two conditions being compared in each case. The result pattern was consistent between the two tests. The one family of comparisons with any statistically significant results was complete-data-versus-imputed data (4 of 9 significant), supporting the conclusion that missingness itself, not the specific mechanism or imputer used to address it, is the dominant, measurable driver of performance loss in this project’s data. Neither the imputer-versus-imputer family nor the mechanism-versus-mechanism family produced a single significant result at n = 5 seeds; this is interpreted throughout the report as "not distinguishable from noise at this sample size," not as evidence of no real difference, since the standard deviation of the complete-data baseline’s own F1 across seeds (0.081) exceeds most of the point-estimate gaps being tested between imputers or mechanisms.

18. Discussion

The project’s central, statistically supported finding is that missingness itself measurably degrades prediction performance relative to complete data, independent of which mechanism produced it or which method was used to address it. This is the single claim in this project backed by a majority-significant test family (4 of 9), and it independently corroborates Shadbahr et al. (2023)’s finding, obtained on different datasets, that missingness rate rather than mechanism or imputer choice is the dominant driver of downstream classifier degradation.

A second notable pattern is the disconnect between reconstruction quality and downstream prediction quality: KNN was consistently the worst imputer by MAE/RMSE, yet was sometimes competitive with or better than the other imputers on downstream F1. A plausible explanation is that Random Forest, as an ensemble of decision trees splitting on threshold values, is comparatively robust to moderate feature-level noise in ways that a direct reconstruction-error metric does not capture; MAE/RMSE and downstream classification accuracy are simply answering different questions about imputation quality.

Third, GA’s demonstrated contribution on this dataset is feature-set stability, not accuracy improvement. The single-seed "GA helps" results observed in the E6 sweep did not survive replication across seeds (Section 16), but the GA-selected feature subset itself - dominated by time, ejection_fraction, and serum_creatinine - was highly stable across five independent runs and, notably, matches Chicco & Jurman (2020)'s independently derived feature-importance ranking for the same dataset, a source the GA process had no access to during its search. This is treated in this report as a genuine positive finding in its own right, even though it is not the finding that a predetermined, in-favour-of-the-proposed-method narrative would have preferred: the project's governing principle explicitly prohibits forcing GA to appear to win, and the evidence here supports "GA as a principled feature-reduction tool" rather than "GA as an accuracy-improving tool" on this dataset.

Fourth, the time feature's outsized predictive contribution must be read as a modelling caveat rather than a clinical discovery. In survival-style clinical data, a short recorded follow-up interval correlates with the death event almost by construction, since a patient who dies is not observed further. time's consistent selection by GA and its central role in the dataset's own origin literature both need to be understood in this light, not presented as an independently surprising predictive insight.

Fifth, the project's binding constraint is sample size, not method choice. Eighteen of the twenty-seven statistical tests conducted were non-significant; given that the complete-data baseline's own across-seed standard deviation in F1 (0.081) exceeds most of the point-estimate gaps under test, the most parsimonious explanation for these non-significant results is that a 299-row dataset and a 60-row held-out test set, even sampled across 5 seeds, does not carry enough statistical power to resolve differences this fine-grained - not that no real difference exists between the compared methods.

19. Existing-Work Comparison

Own results are compared against selected published literature contextually, without presenting numbers computed on different datasets as a direct head-to-head result. This project's RQ1 finding - that missingness rate matters more than mechanism - agrees with Shadbahr et al. (2023). Its RQ2 finding - that no single imputer statistically dominates - agrees with both Ren et al. (2024)'s systematic review of EHR imputation methods and Aracri et al. (2025)'s comparison for dementia classification. Its RQ3 finding on feature-selection stability agrees, independently, with Chicco & Jurman (2020)'s feature-importance analysis for the same underlying dataset. Its RQ3 finding on accuracy does not agree with Kumar & Sahoo (2017), who reported a favourable GA-plus-Naive- Bayes result on a different cardiovascular dataset; a plausible, though unproven, explanation offered here is that this project's small feature space (12 features) and training set (239 rows) give a genetic wrapper search comparatively little room to find a genuinely superior subset, and comparatively more room to overfit its own cross-validation-based fitness signal - a known general risk of wrapper-based feature selection on small, noisy datasets (Pudjihartono et al., 2022). This divergence is reported directly, not omitted, consistent with the project's governing principle that evidence should not be forced toward a predetermined conclusion.

20. Limitations

- The study uses a single primary dataset of 299 rows (239 train / 60 test), small by general machine-learning standards; this is an intentional scope-control decision restricting the main study to one primary dataset within a single-semester timeframe.

- Five seeds is a statistically thin repeated-measures design: 18 of 27 paired tests are non-significant, which should be read as "not distinguishable from noise at this sample size," not as "no real effect exists."

- No multiple-comparison correction was applied to the 27 statistical tests; at n = 5, any standard correction would make almost every comparison non-significant regardless of true effect size, so individual p-values are read as exploratory/suggestive rather than confirmatory.

- Each seed reseeds the entire pipeline, including the train/test split itself, not only model training - a more thorough but higher-variance multi-seed design than reseeding the model alone;

this should be kept in mind when comparing this project's variance estimates to studies that reseed only the model.

- E6 (Imputation + GA + Random Forest) multi-seed coverage is a 3-cell/3-seed subset, not the full 9-cell/5-seed grid, for compute-budget reasons; sufficient to show that the single-seed GA pattern does not replicate, but not an exhaustive test of every cell.

- GA fitness evaluation used 50 trees against the final model's 200-tree configuration, a documented roughly five-times cost reduction for the search process only; final reported comparisons use 200 trees on both sides, so reported performance numbers are unaffected, though the GA search itself operated on a noisier fitness signal than the final evaluation.

- KNN's k = 5 and IterativeImputer's settings are scikit-learn defaults, not cross-validation-tuned; neither choice is unreasonable, but neither is optimized.

- Missingness and imputation are scoped to the seven continuous features; the five binary clinical flags are not masked, since mean/median imputation is not a meaningful reconstruction target for a binary indicator.

- The time column's leakage-adjacency, discussed in Section 18 (Discussion) and Section 7 (Dataset), is retained as a documented methodological decision rather than removed, and its outsized influence on Random Forest performance should not be read as a genuine predictive discovery.

- Two literature entries - Kumar & Sahoo (2017) and Yaqoob et al. (2025) - carry a verification caveat: their full text could not be retrieved due to publisher access restrictions, so bibliographic details and reported findings are drawn from indexed metadata rather than a direct primary-source read. One entry, Grzesiak et al. (2025), is an unreviewed arXiv preprint and is labelled accordingly.

- No single master run_all.py execution script exists; the project instead uses one independently re-runnable script per experiment group, a documented scope-control simplification.

21. Future Work

- Additional seeds beyond the current five, to resolve the several near-threshold p-values observed in the statistical tests and give RQ1 and RQ2 more statistical power.

- Full 9-cell/5-seed E6 coverage, extending the current 3-cell/3-seed replication subset, to confirm across every mechanism-by-level cell (not only the three sampled) that GA does not reliably improve accuracy once imputation is already in the pipeline.

- A second dataset for external validation, most plausibly Chronic Kidney Disease (ranked second among the candidates in docs/dataset_selection.md), whose genuine natural missingness would let RQ1 be tested against real rather than only synthetic missingness.

- Cross-validation-tuned KNN k and IterativeImputer settings, rather than scikit-learn defaults, to check whether the disconnect between reconstruction quality and downstream prediction quality (Section 20) persists under tuning.

- A hybrid filter-then-wrapper feature-selection approach (Pudjihartono et al., 2022), to test whether GA's plausible overfitting risk on a small cross-validation signal (offered here as the explanation for the project's mixed RQ3 result) is reduced by pre-filtering the candidate feature space before the genetic search begins.

- An explicit sensitivity analysis on the time feature, rerunning the project's headline comparisons with time excluded, to directly quantify how much of the reported Random Forest performance depends on this leakage-adjacent feature.

- An optional Autoencoder-based imputer, only if a future, substantially larger dataset or a specific reviewer request justifies the added complexity; not pursued here, since the classical and iterative imputers already implemented are sufficient to answer RQ1 and RQ2 at this dataset's scale.

22. Conclusion

This project set out to determine how missing data affects clinical prediction, which imputation method performs best, and whether GA-based feature selection improves Random Forest prediction, using a controlled, repeated-seed experimental design on the Heart Failure Clinical Records dataset. The evidence supports one clear, statistically defensible conclusion: missingness itself, independent of its mechanism or the method used to address it, measurably reduces prediction performance. No imputation method and no missingness mechanism could be statistically distinguished from its alternatives at this dataset's sample size, so no ranking among mean, median, KNN, and IterativeImputer, nor among MCAR, MAR, and MNAR, is claimed. GA-based feature selection reliably and reproducibly identified a small, clinically plausible feature subset that independently matches this dataset's own origin literature, but did not reliably improve prediction accuracy once tested across multiple seeds rather than a single split - a genuine, non-forced finding that GA's demonstrated value here is principled dimensionality reduction rather than an accuracy gain. Consistent with the project's governing principle of not forcing a predetermined conclusion, these results - including the non-significant comparisons and GA's mixed accuracy result - are reported as the actual, replicated evidence produced by this study, and are offered as a defensible, honestly limited research result rather than a claim of a superior proposed method.

References

Full bibliographic detail with DOIs is provided in report_data/references.bib and results/literature_matrix.csv; the fifteen references cited above are, in the order first cited: Rubin (1976); van Buuren & Groothuis-Oudshoorn (2011); Stekhoven & Buhlmann (2012); Troyanskaya et al. (2001); Breiman (2001); Leardi, Boggia, & Terrile (1992); Demsar (2006); Shadbahr et al. (2023); Ren et al. (2024); Aracri et al. (2025); Chicco & Jurman (2020); Kumar & Sahoo (2017); Pudjihartono et al. (2022); Yaqoob et al. (2025); Grzesiak et al. (2025).